

B.Math.(Hons.) IIIrd year
Semestral examination
7th May 2012
Algebraic Number Theory
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Q 1. In a Dedekind domain, prove that an ideal divides another ideal if and only if it contains it.

OR

Let $L \supset K$ be number fields. Prove that only finitely many prime ideals of O_K can ramify in O_L .

Q 2. If $[K : \mathbf{Q}] = 3$ satisfies $-49 < \text{disc}(K) < 0$, show that the class group of K is trivial. You may use Minkowski's bound $\frac{n!}{n^n} \left(\frac{4}{\pi}\right)^{r_2} \sqrt{|\text{disc}(K)|}$.

OR

Using the fact that $K := \mathbf{Q}(\sqrt{-23})$ is contained in the field $L := \mathbf{Q}(\zeta_{23})$ and that O_K contains non-principal ideals, prove that O_L also contains non-principal ideals.

Q 3. Determine the fundamental unit of $\mathbf{Q}(\sqrt{d^2 - 1})$ where $d > 1$.

OR

For a positive integer n , prove that $\frac{\sin(k\pi/n)}{\sin(\pi/n)}$ is a unit in $\mathbf{Q}(\zeta_n)$ for $(k, n) = 1$.

Q 4. Let $K = \mathbf{Q}(\alpha)$ be a number field and let $L = \mathbf{Q}(\alpha_1, \dots, \alpha_n)$ where the minimal polynomial of α is $\prod_{i=1}^n (X - \alpha_i)$. If a prime number p ramifies in O_L , prove that it ramifies in O_K also.

OR

If L/K is a Galois extension of number fields, and if P is a maximal ideal of O_K , then prove that the number of prime ideals of O_L lying over P , divides $[L : K]$.

Q 5. Let p be a prime and let n be a positive integer. Consider the cyclotomic polynomial Φ_{p^n+1} whose roots are the primitive $(p^n + 1)$ -th roots of unity. Determine the number of irreducible factors of the polynomial $\Phi_{p^n+1}(X) \bmod p$ and their degrees.

Q 6. Let p be any odd prime $\equiv 1 \pmod{4}$. Let a, b be positive integers such that $a^2 + b^2 \equiv -1 \pmod{p}$. Consider the lattice Γ in $\mathbf{R}^4$ with basis vectors $v_1 = (p, 0, 0, 0), v_2 = (0, p, 0, 0), v_3 = (a, b, 1, 0), v_4 = (b, -a, 0, 1)$ and the open ball in $\mathbf{R}^4$ centred at the origin and radius $\sqrt{2p}$. Use Minkowski's lemma to deduce that p is a sum of four squares of integers.